AutoSave Off | Complete Solved - Dashboard - Superstore Data.xlsx | Search | Rahim Zulfiqar Ali

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A5

# Excel Basement | Sales Analytical Dashboard

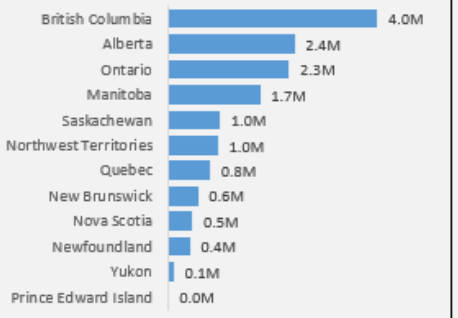
Time Intelligence Filters

2009 2010 2011 2012 | Q1 Q2 Q3 Q4 | Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

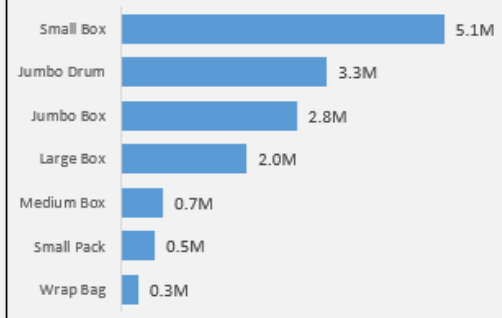
## Quick Summary | Key Performance Indicators

| Total Orders $ | Net Orders $ | Returns $ | Net Sell Price $ | Shipping Amount $ | Count Products |
|---|---|---|---|---|---|
| 14.7M | 13.1M | (1.6M) | 713K | 108K | 1,263 |

### Total Orders $ by Province



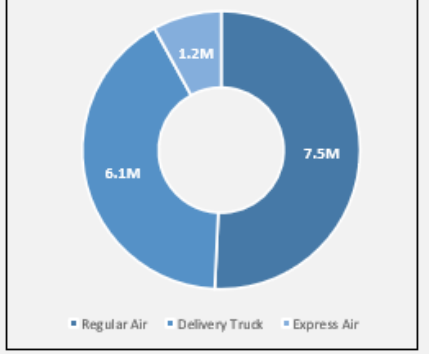

### Total Orders $ by Product Container




### Total Orders $ by Customer Segment



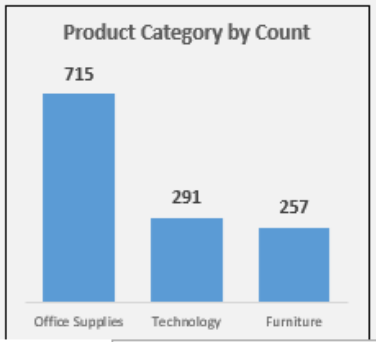

### Total Orders $ by Ship Mode




### Order Amount by Product Category



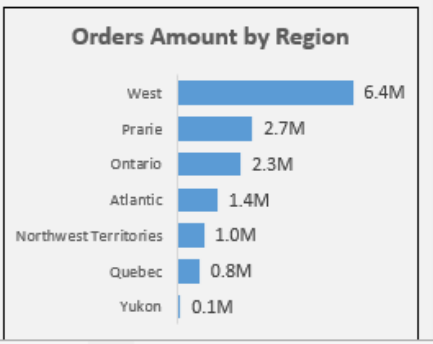

### Product Category by Count




### Total Order $ by Month Name




### Orders Amount by Region



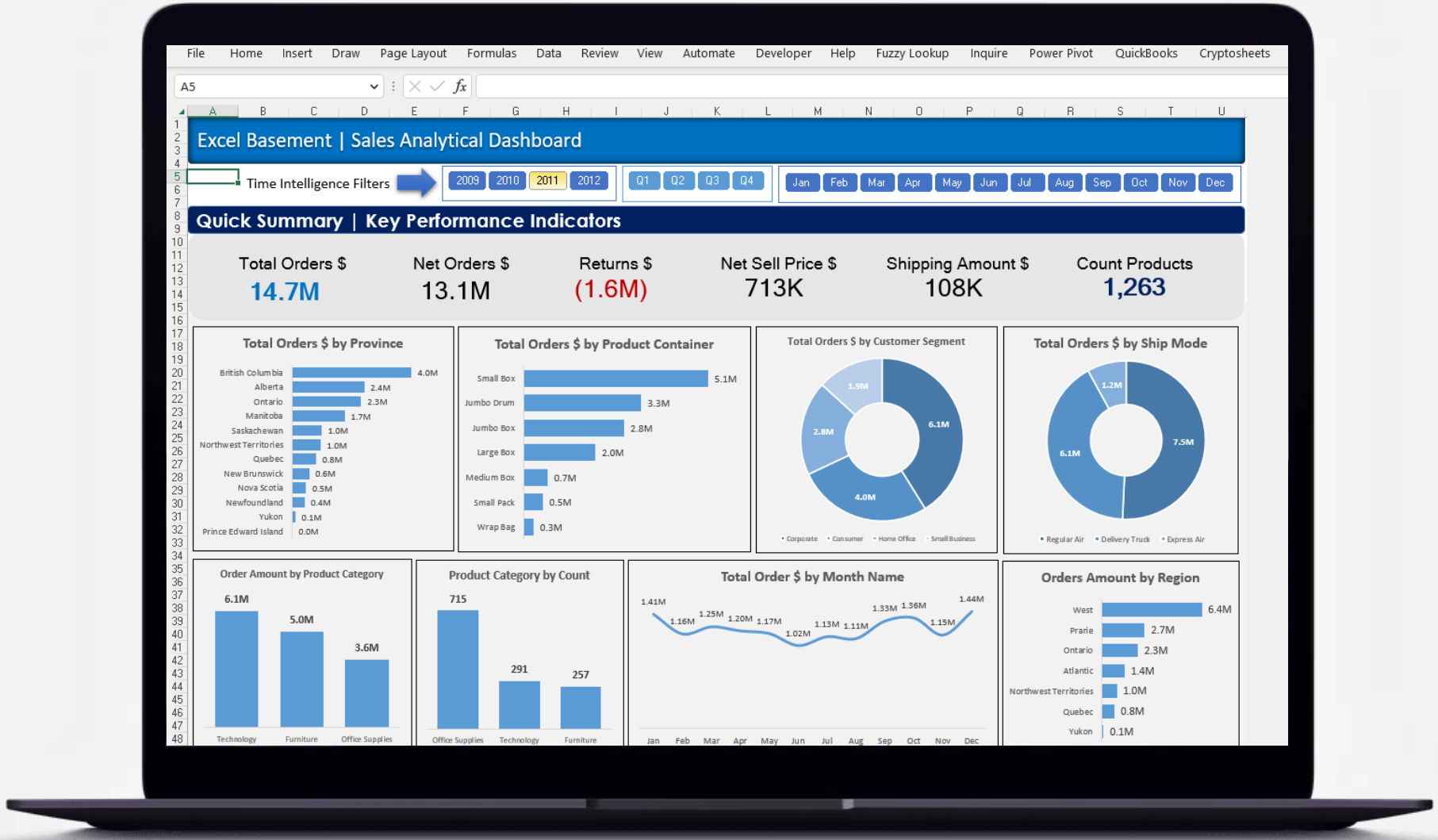

Dates | Workings | Dashboard

Ready | Accessibility: Investigate



**Advanced Sales Analytical Dashboard using Data Analysis Expressions (DAX) & Visualizations Rules + Tips in Microsoft Excel**

# Dashboard Preview



# What is DAX?

Data Analysis Expressions is the native formula and query language for Microsoft PowerPivot, Power BI Desktop and SQL Server Analysis Services Tabular models. **Wikipedia**

Data Analysis Expressions (DAX) is a formula expression language used in Analysis Services, Power BI, and Power Pivot in Excel. DAX formulas include functions, operators, and values to perform advanced calculations and queries on data in related tables and columns in tabular data models. **Microsoft Learn**

## In More Simple Words…. DAX!

DAX (Data Analysis Expressions) is a formula language used in Microsoft PowerBI and Excel to create custom calculations and aggregations in a data model.

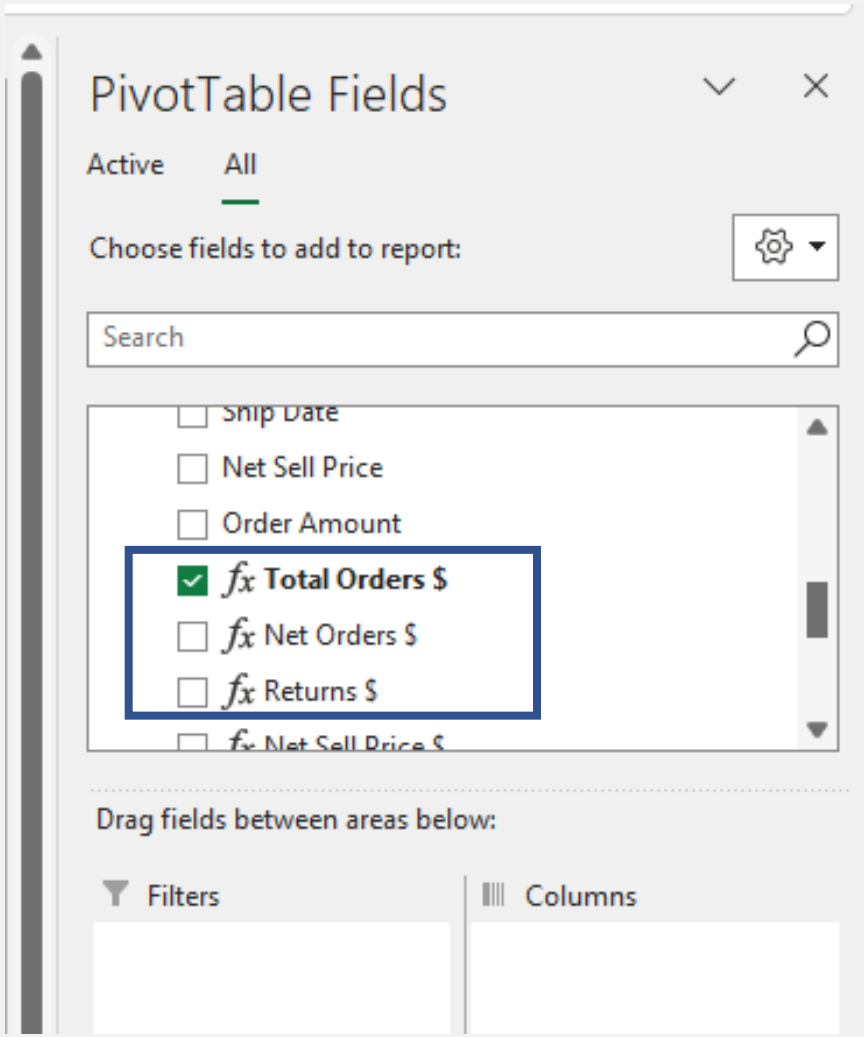
It allows you to perform data analysis, data modeling, and data manipulation tasks, such as creating calculated columns, calculated tables, and measures.

DAX expressions are used to perform operations on the data, and can be based on simple mathematical operations, or more complex logic.

# What is Measure?

Measures are **calculations used in data analysis**. Examples commonly found in business reports include sums, averages, minimum or maximum values, counts, or more advanced calculations that you create using a Data Analysis Expressions (DAX) formula.

You can name measures whatever you want and add them to a new or existing visualization just like any other field.



# MEASURE – Implicit vs. Explicit

## Implicit Measures

- Automatically defined for all numeric columns
- Created measure “Sum of <Field>”
- Can be changed to Avg, Count, Min, Max

## Explicit Measures

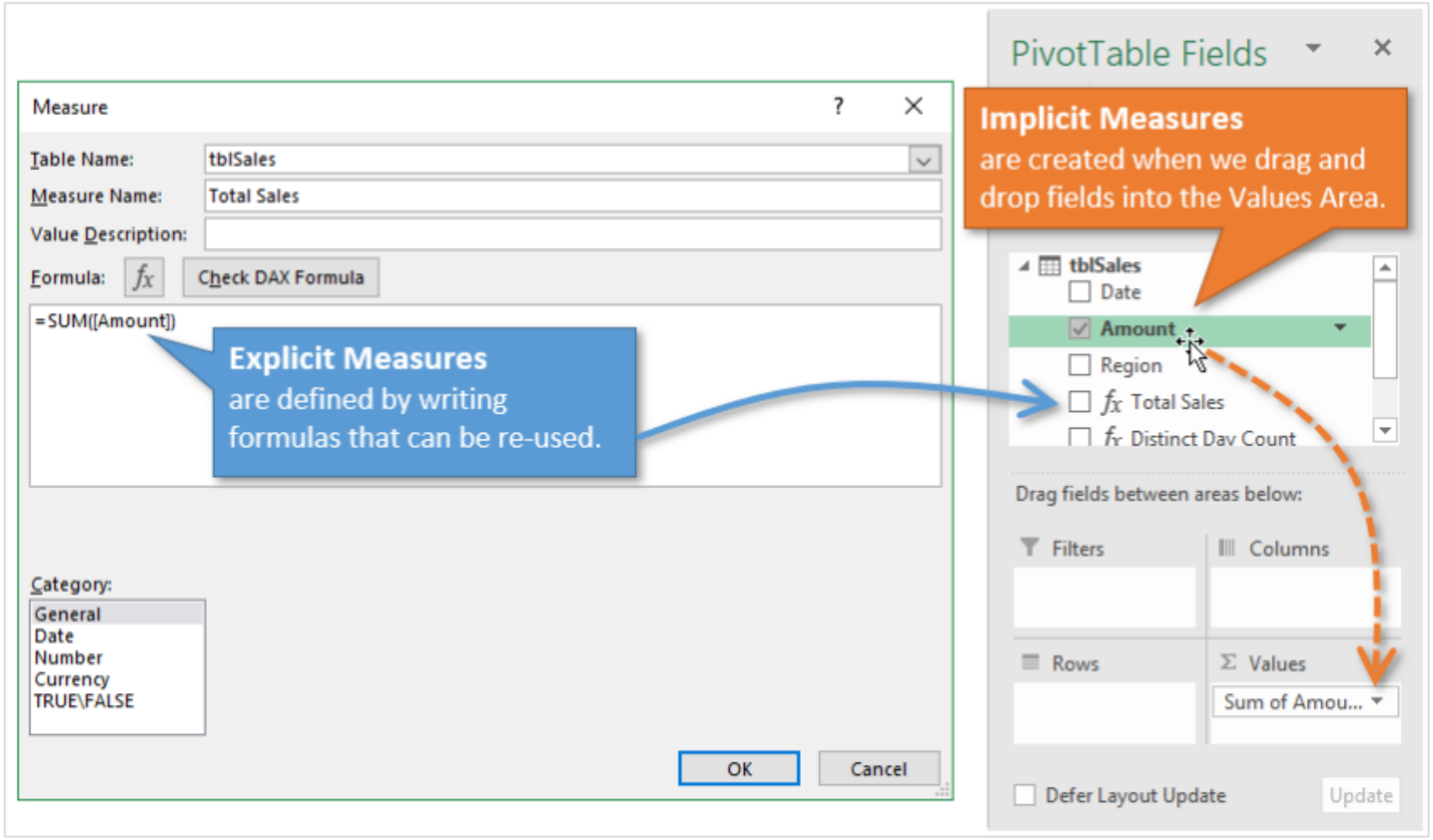
- Gives a name to a DAX expression (formula)
- Can use complex expressions
- Interaction with Filter Context
- [Margin %] =SUM([Margin])/SUM([Amount])

## **Explicit measures = more flexibility**!

While implicit measures can support some basic scenarios, as soon as your report needs more complex calculations, you'll have to switch to explicit measures.

However, the main advantage of using explicit measures instead of implicit ones is their reusability.

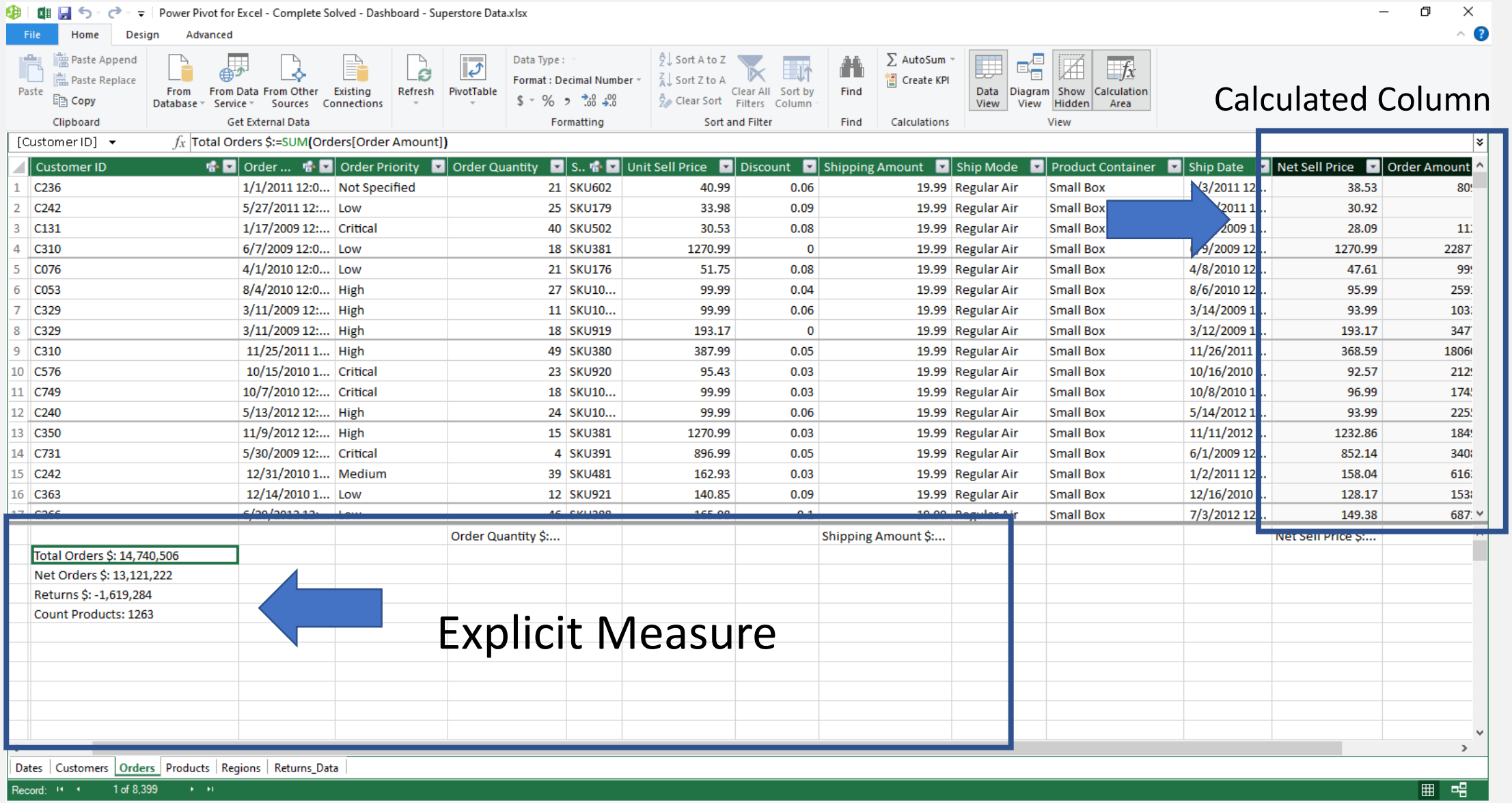
# MEASURE – Implicit vs. Explicit



# Calculated Column & Measure in Excel | DAX

A **calculated column** belongs to a ***single table***, while a **measure** belongs to the ***whole data model.***

A **calculated column** is evaluated in a **row context** (row by row, like in an excel table), while a **measure** is evaluated in the **filter context.**


Power Pivot for Excel - Complete Solved - Dashboard - Superstore Data.xlsx
File
Home
Design
Advanced
Paste
Paste Append
Paste Replace
Copy
Clipboard
From Database
From Data Service
From Other Sources
Existing Connections
Get External Data
Refresh
PivotTable
Data Type :
Format : Decimal Number
Formatting
Sort A to Z
Sort Z to A
Clear Sort
Clear All Filters
Sort by Column
Sort and Filter
Find
AutoSum
Create KPI
Calculations
Data View
Diagram View
Show Hidden
Calculation Area
View
Calculated Column
[Customer ID]
Total Orders $:=SUM(Orders[Order Amount])
Customer ID
Order ...
Order Priority
Order Quantity
S..
Unit Sell Price
Discount
Shipping Amount
Ship Mode
Product Container
Ship Date
Net Sell Price
Order Amount
C236 1/1/2011 12:0... Not Specified 21 SKU602 40.99 0.06 19.99 Regular Air Small Box 38.53
C242 5/27/2011 12:... Low 25 SKU179 33.98 0.09 19.99 Regular Air Small Box 30.92
C131 1/17/2009 12:... Critical 40 SKU502 30.53 0.08 19.99 Regular Air Small Box 28.09
C310 6/7/2009 12:0... Low 18 SKU381 1270.99 0 19.99 Regular Air Small Box 1270.99
C076 4/1/2010 12:0... Low 21 SKU176 51.75 0.08 19.99 Regular Air Small Box 4/8/2010 12.. 47.61
C053 8/4/2010 12:0... High 27 SKU10... 99.99 0.04 19.99 Regular Air Small Box 8/6/2010 12.. 95.99
C329 3/11/2009 12:... High 11 SKU10... 99.99 0.06 19.99 Regular Air Small Box 3/14/2009 1.. 93.99
C329 3/11/2009 12:... High 18 SKU919 193.17 0 19.99 Regular Air Small Box 3/12/2009 1.. 193.17
C310 11/25/2011 1... High 49 SKU380 387.99 0.05 19.99 Regular Air Small Box 11/26/2011 .. 368.59
C576 10/15/2010 1... Critical 23 SKU920 95.43 0.03 19.99 Regular Air Small Box 10/16/2010 .. 92.57
C749 10/7/2010 12:... Critical 18 SKU10... 99.99 0.03 19.99 Regular Air Small Box 10/8/2010 1.. 96.99
C240 5/13/2012 12:... High 24 SKU10... 99.99 0.06 19.99 Regular Air Small Box 5/14/2012 1.. 93.99
C350 11/9/2012 12:... High 15 SKU381 1270.99 0.03 19.99 Regular Air Small Box 11/11/2012 .. 1232.86
C731 5/30/2009 12:... Critical 4 SKU391 896.99 0.05 19.99 Regular Air Small Box 6/1/2009 12.. 852.14
C242 12/31/2010 1... Medium 39 SKU481 162.93 0.03 19.99 Regular Air Small Box 1/2/2011 12.. 158.04
C363 12/14/2010 1... Low 12 SKU921 140.85 0.09 19.99 Regular Air Small Box 12/16/2010 .. 128.17
Small Box 7/3/2012 12.. 149.38
Order Quantity $:...
Shipping Amount $:...
Net Sell Price $:...
Total Orders $: 14,740,506
Net Orders $: 13,121,222
Returns $: -1,619,284
Count Products: 1263
Explicit Measure
Dates
Customers
Orders
Products
Regions
Returns_Data
Record:
1 of 8,399

# Star schema overview

Star schema is a mature modeling approach widely adopted by relational data warehouses. It requires modelers to classify their model tables as either dimension or fact.

**Dimension tables** describe business entities—the things you model. Entities can include products, people, places, and concepts including time itself.

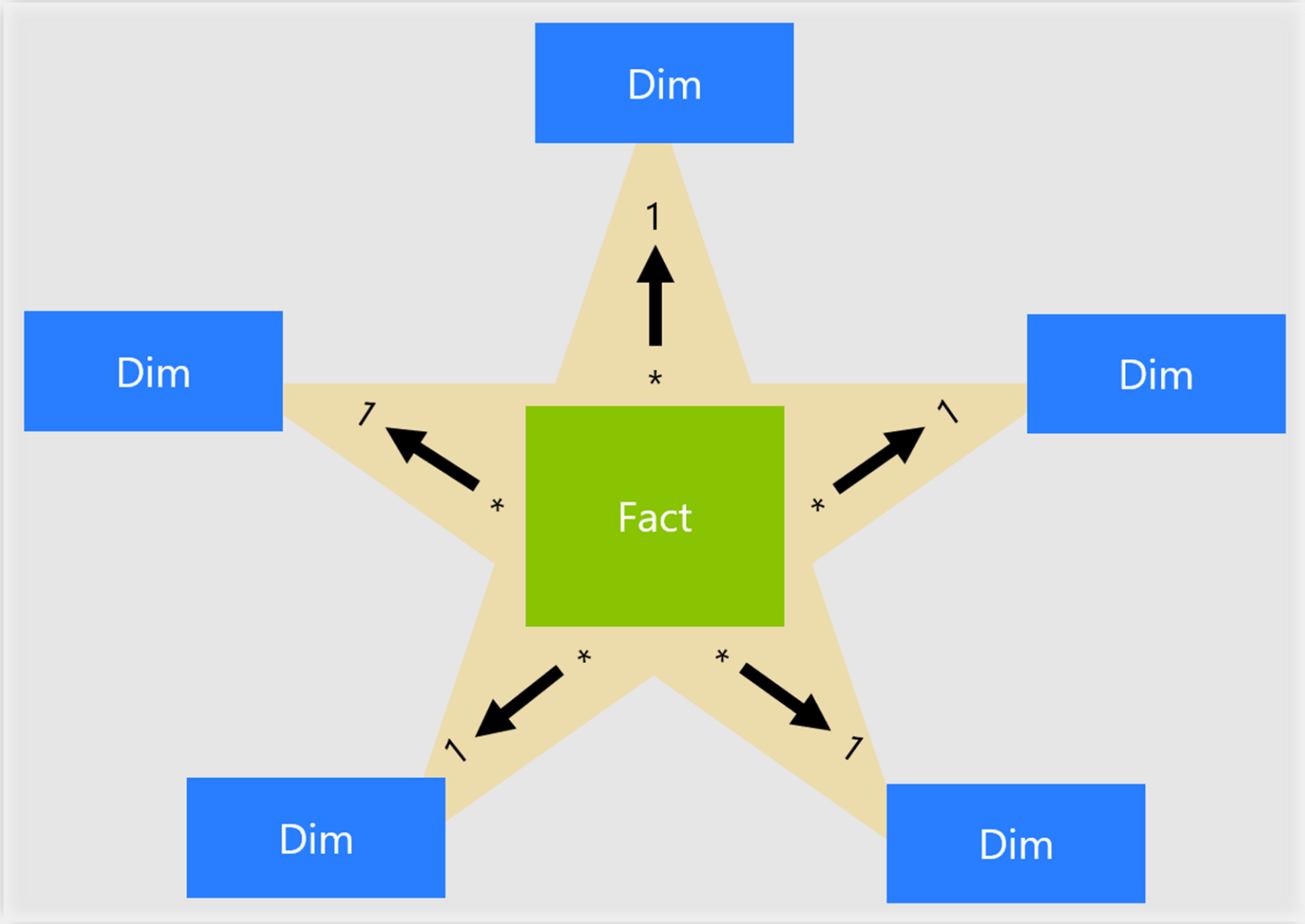
The most consistent table you'll find in a star schema is a date dimension table. A dimension table contains a key column (or columns) that acts as a unique identifier, and descriptive columns

**Fact tables** store observations or events, and can be sales orders, stock balances, exchange rates, temperatures, etc. A fact table contains dimension key columns that relate to dimension tables, and numeric measure columns.

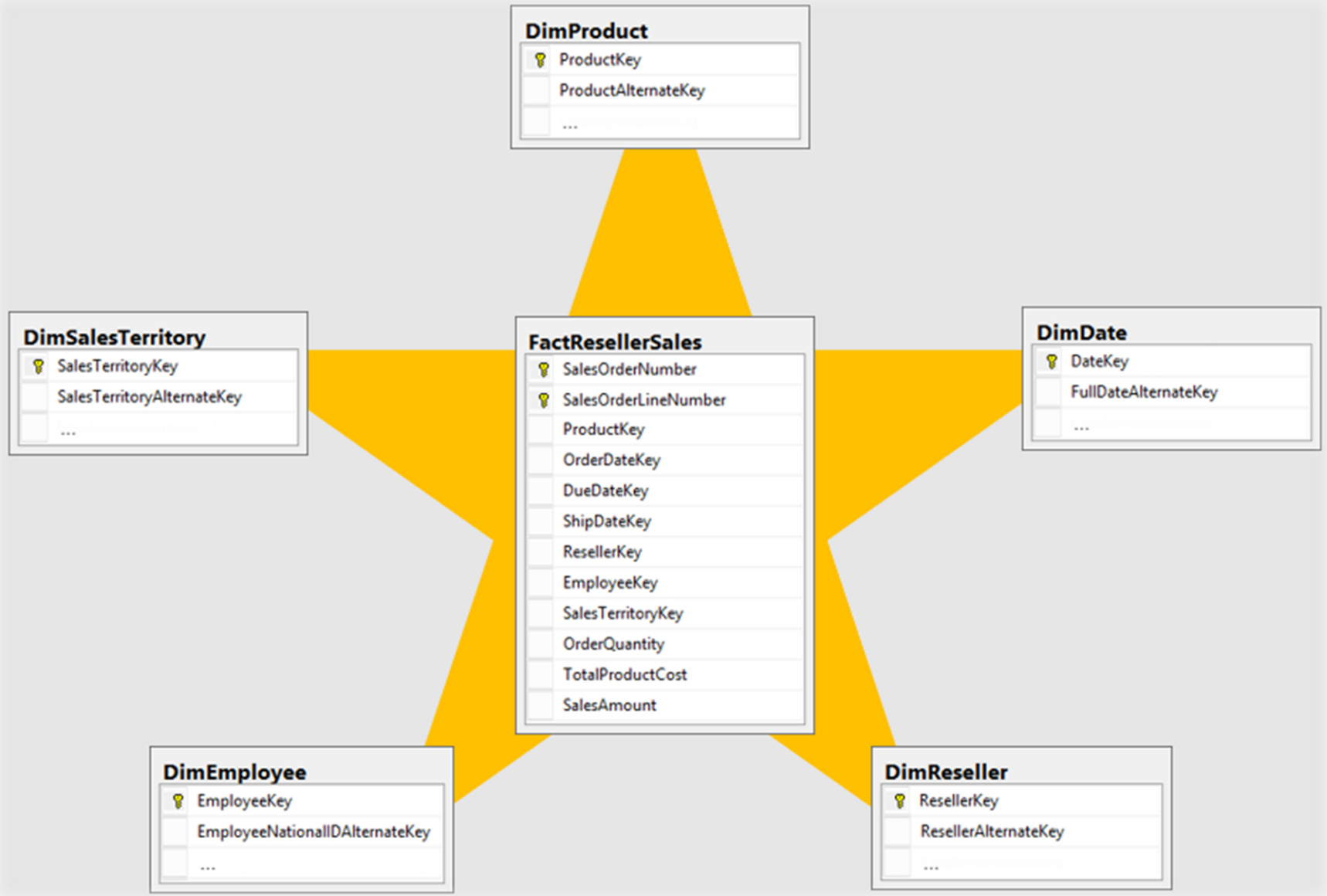
The **dimension** key columns determine the dimensionality of a fact table, while the dimension key values determine the granularity of a fact table.

| | Dimension | Fact |
|---|---|---|
| **Model Purpose** | • Defines business entities ("things")<br>• For example: Date, products, people, places, and concepts | • Stores observations or events<br>• For example: Sale orders, stock balances, exchange rates, temperatures, etc. |
| **Table Structure** | • A single dimension key column (unique identifier)<br>• Descriptive columns (attributes) | • Dimension key columns<br>• Numeric measure columns |
| **Data** | • Typically, contains a (relatively) small number of rows | • Can contain a very large number of rows<br>• Must be loaded at a consistent grain (level of detail) |
| **Query Purpose** | • To **filter**<br>• To **group** | • To **summarize** |

# STAR SCHEMA



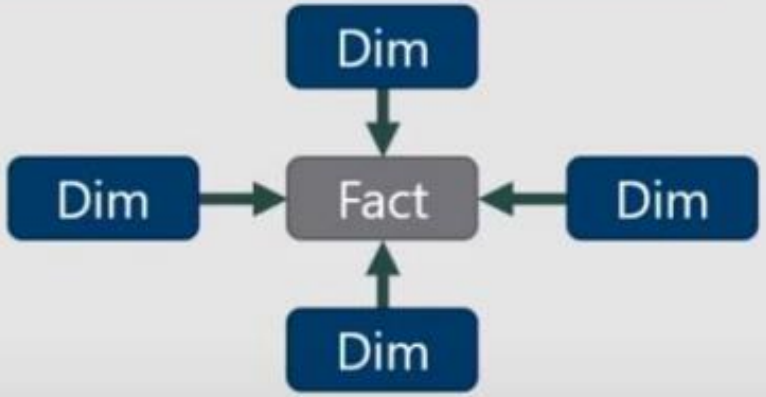
# STAR SCHEMA


DimProduct
ProductKey
ProductAlternateKey
...
DimSalesTerritory
SalesTerritoryKey
SalesTerritoryAlternateKey
...
FactResellerSales
SalesOrderNumber
SalesOrderLineNumber
ProductKey
OrderDateKey
DueDateKey
ShipDateKey
ResellerKey
EmployeeKey
SalesTerritoryKey
OrderQuantity
TotalProductCost
SalesAmount
DimDate
DateKey
FullDateAlternateKey
...
DimEmployee
EmployeeKey
EmployeeNationalIDAlternateKey
...
DimReseller
ResellerKey
ResellerAlternateKey
...

# Relationship Schemas

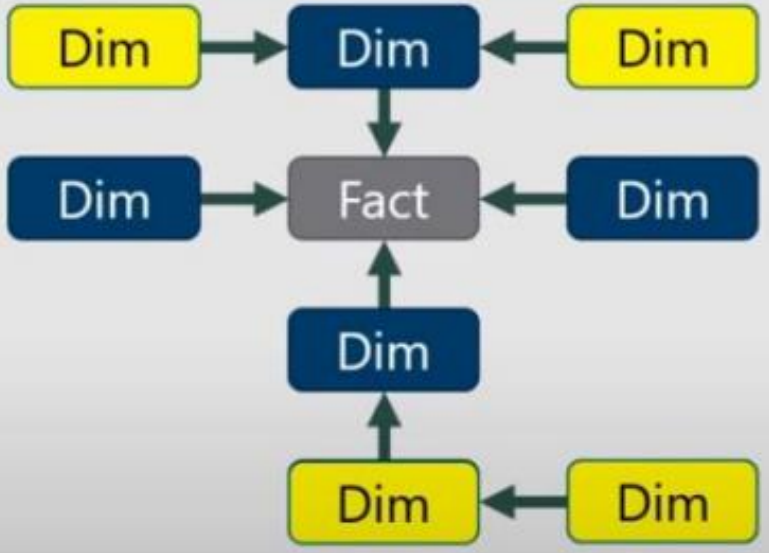
**Star**

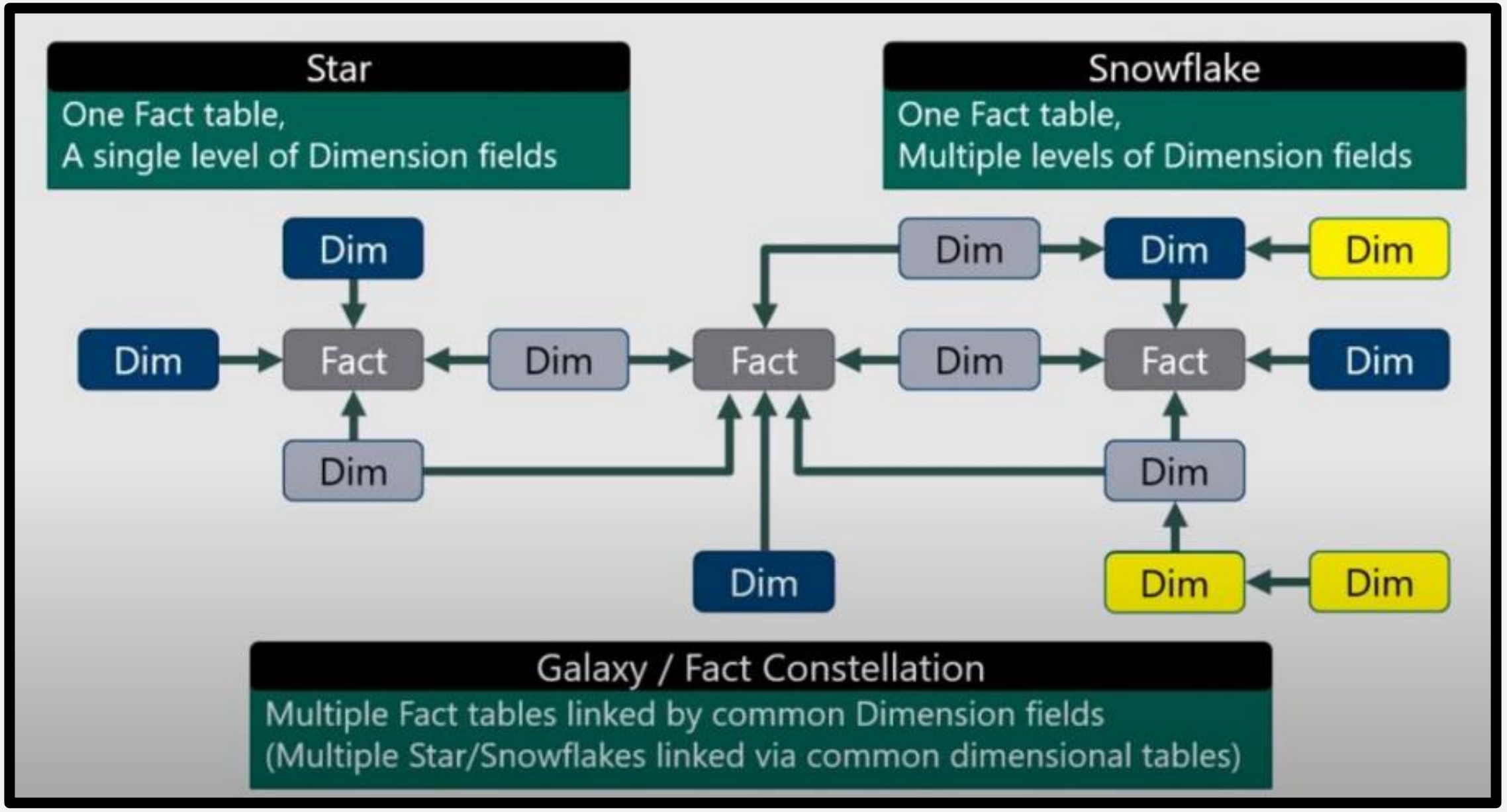
One Fact table,
A single level of Dimension fields




**Snowflake**

One Fact table,
Multiple levels of Dimension fields




Star
One Fact table,
A single level of Dimension fields
Snowflake
One Fact table,
Multiple levels of Dimension fields
Dim
Dim
Fact
Dim
Dim
Fact
Dim
Dim
Fact
Dim
Dim
Dim
Dim
Dim
Dim
Dim
Dim
Galaxy / Fact Constellation
Multiple Fact tables linked by common Dimension fields
(Multiple Star/Snowflakes linked via common dimensional tables)

# Primary Key & Foreign Key (Basic concept)

| studentId | firstName | lastName | courseId |
|---|---|---|---|
| L0002345 | Jim | Black | C002 |
| L0001254 | James | Harradine | A004 |
| L0002349 | Amanda | Holland | C002 |
| L0001198 | Simon | McCloud | S042 |

Foreign Key

Fact Table

Real time transactions & Duplicate Records

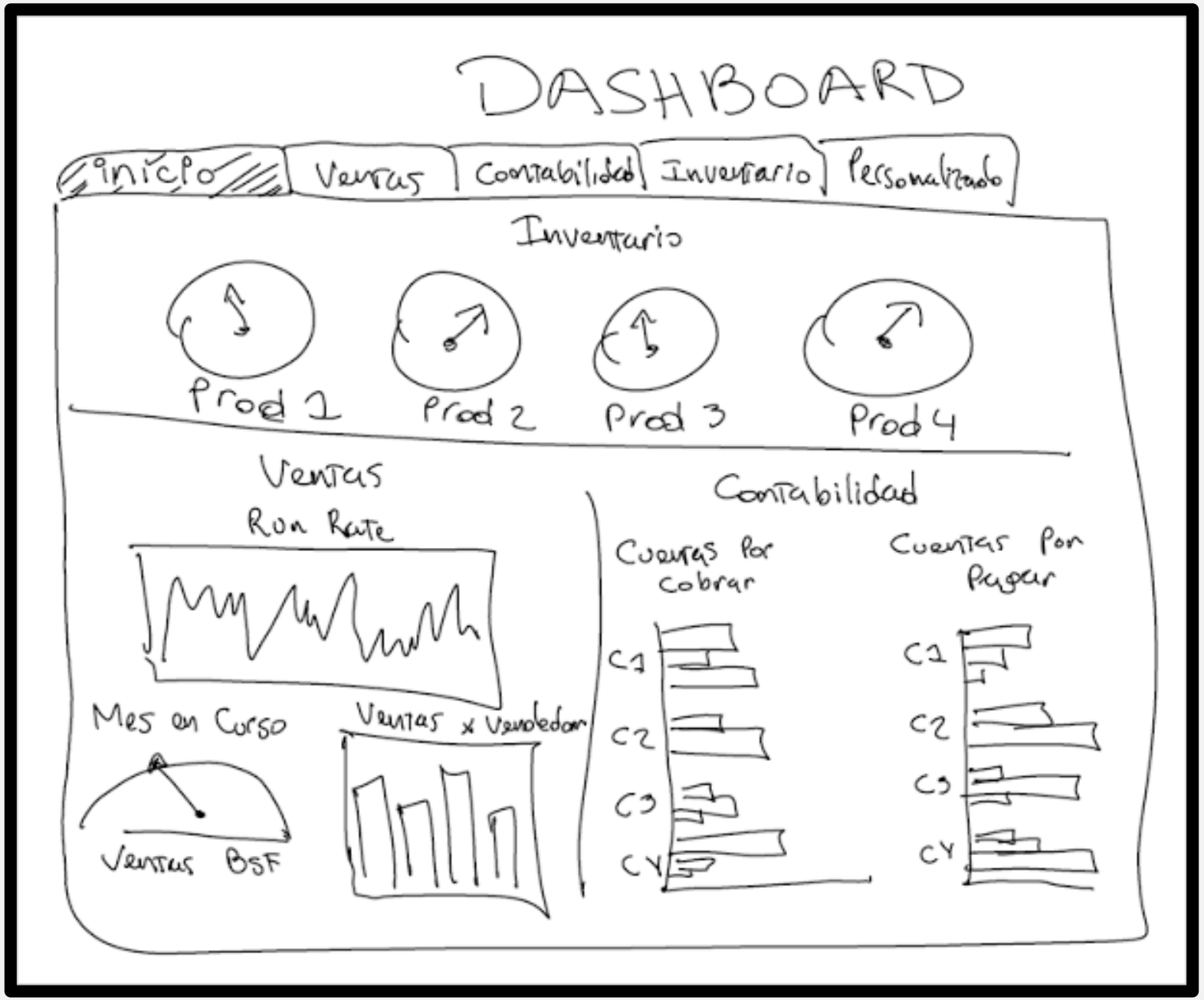
Relationship

Primary Key

Dimension Table

Unique Records

| courseId | courseName |
|---|---|
| A004 | Accounts |
| C002 | Computing |
| P301 | History |
| S042 | Short Course |


DASHBOARD
Inicio
Ventas
Contabilidad
Inventario
Personalizado
Inventario
Prod 1
Prod 2
Prod 3
Prod 4
Ventas
Run Rate
Mes en Curso
Ventas BSF
Ventas x Vendedor
Contabilidad
Cuentas por cobrar
C1
C2
C3
CY
Cuentas por Pagar
C1
C2
C3
CY

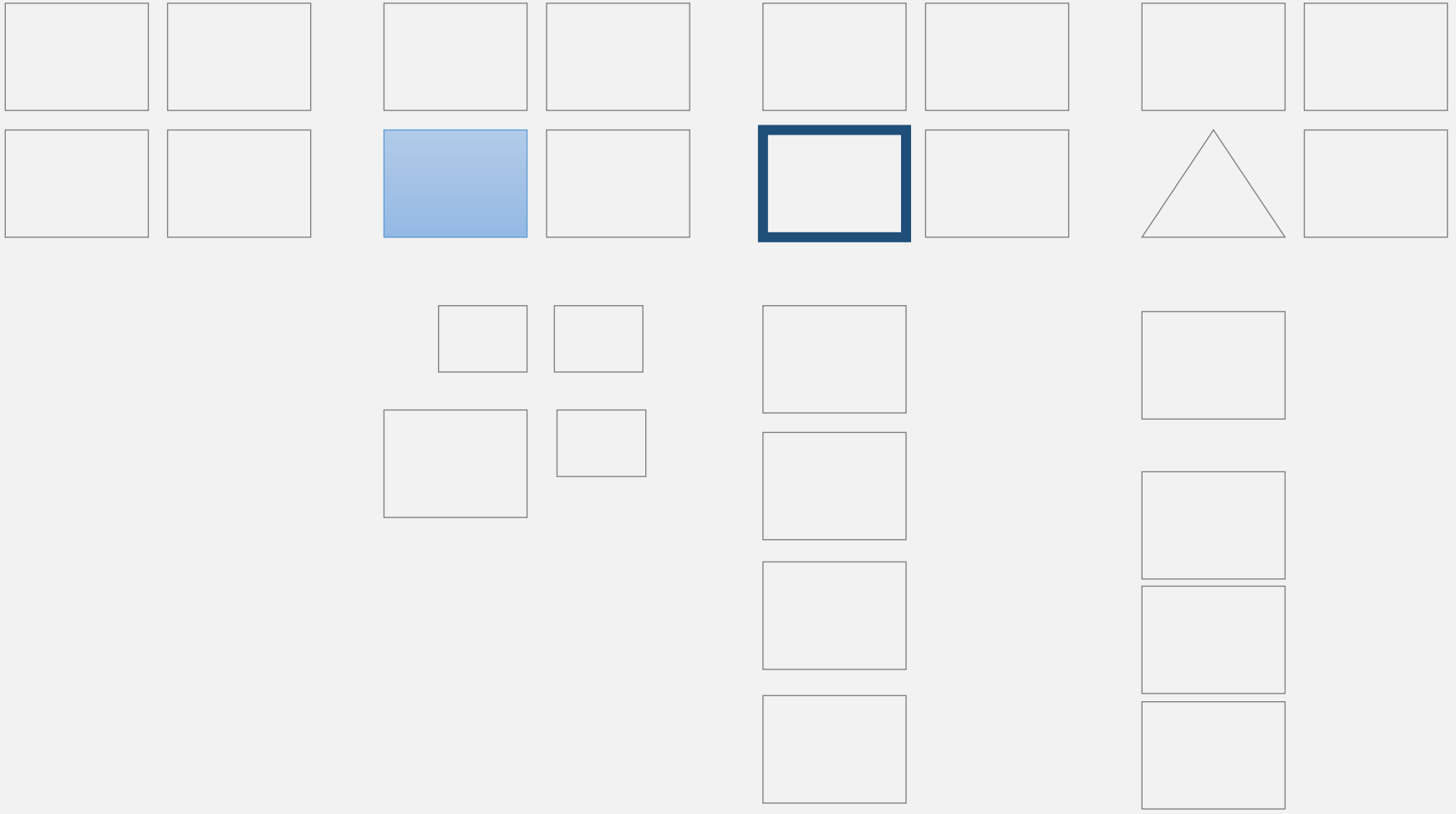
# C.R.A.P.,

a design principle developed by Robin Patricia Williams, stands for Contrast, Repetition, Alignment, and Proximity.

By leveraging CRAP, you can consistently deliver effective designs, whether it's for a website, a landing page, a checkout page, an eBook, or just a banner ad.

# CRAP Rule

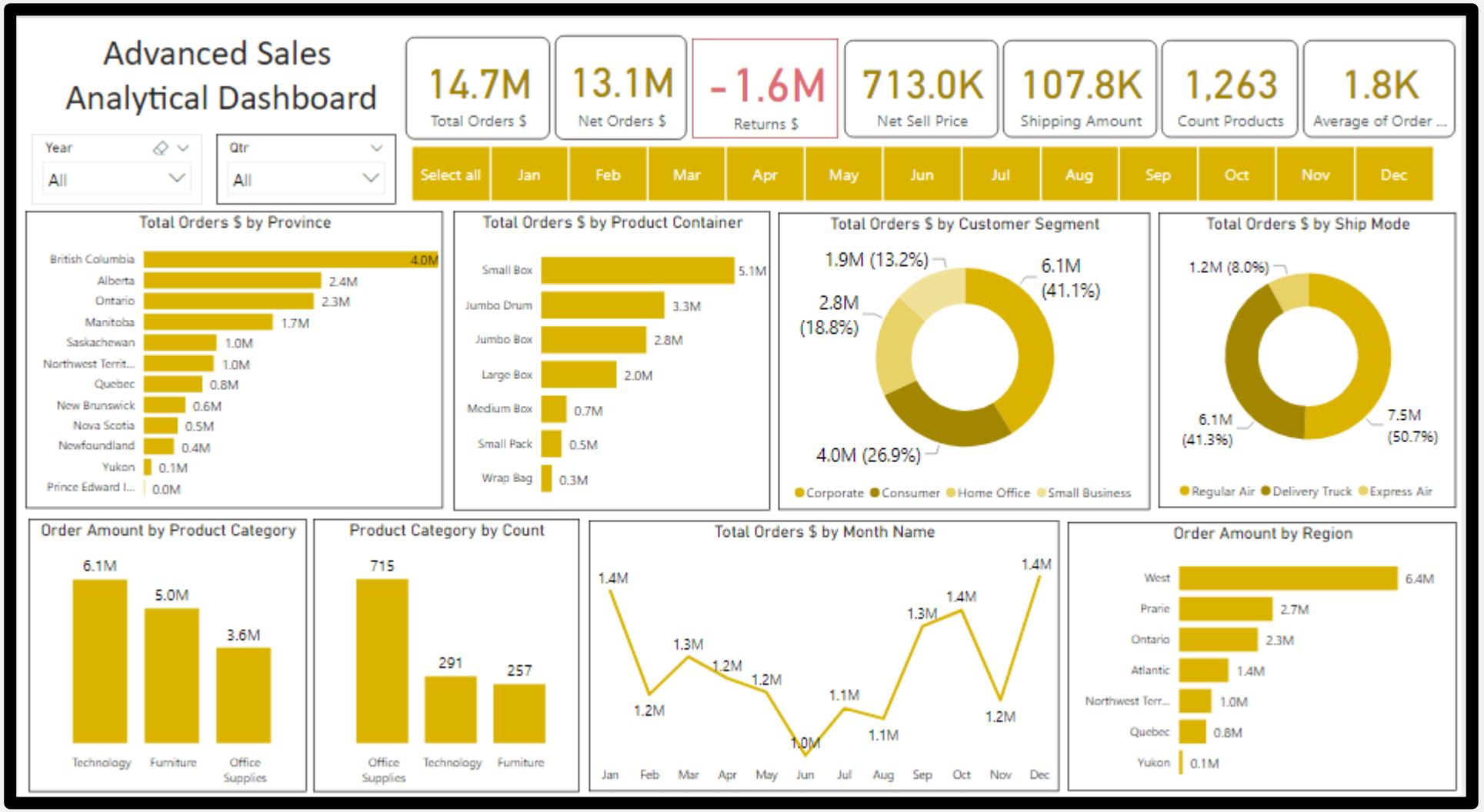
# Contrast



# Contrast by

- Color
- Thickness
- Shape
- Size
- Position
- Distance

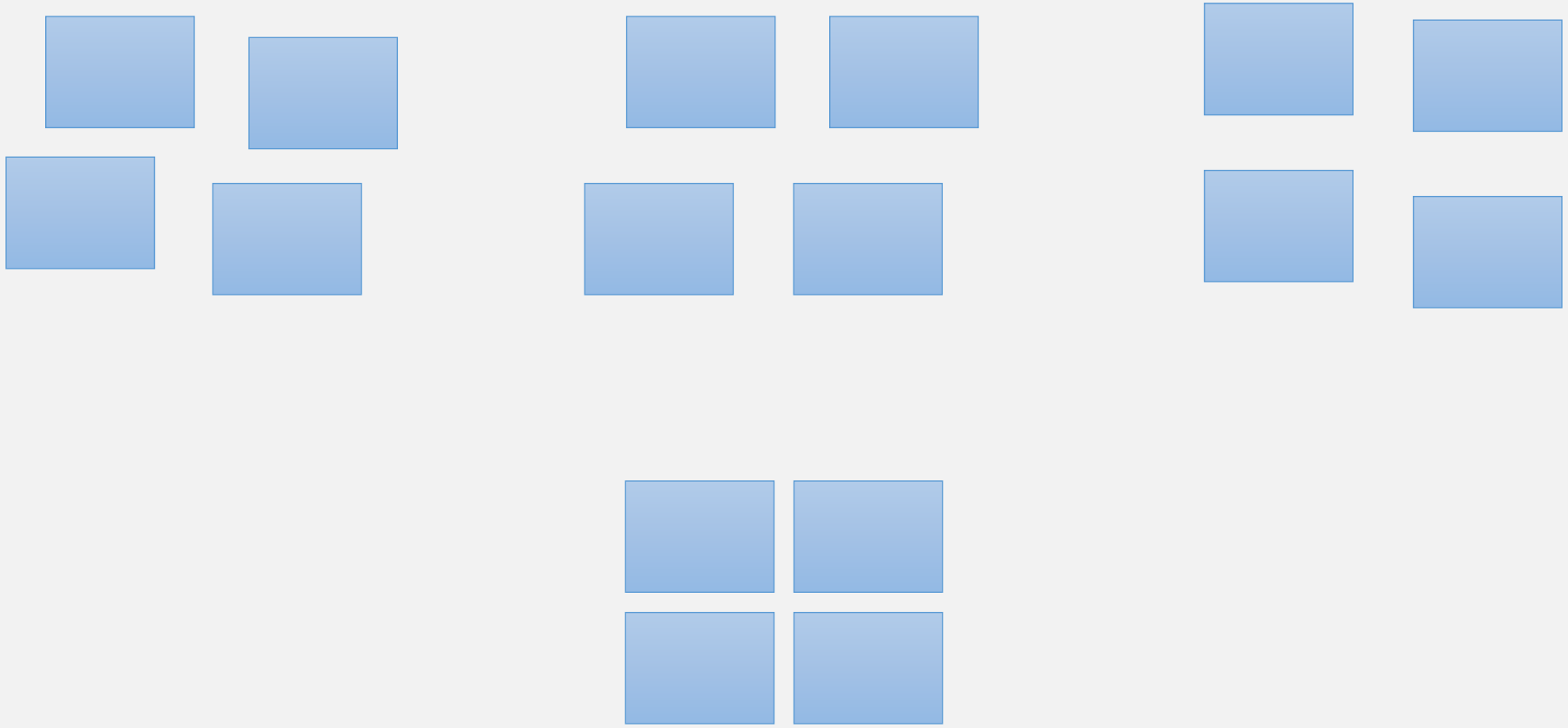
# Repetition


Advanced Sales Analytical Dashboard
14.7M
Total Orders $
13.1M
Net Orders $
-1.6M
Returns $
713.0K
Net Sell Price
107.8K
Shipping Amount
1,263
Count Products
1.8K
Average of Order ...
Year
All
Qtr
All
Select all
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Total Orders $ by Province
British Columbia 4.0M
Alberta 2.4M
Ontario 2.3M
Manitoba 1.7M
Saskachewan 1.0M
Northwest Territ... 1.0M
Quebec 0.8M
New Brunswick 0.6M
Nova Scotia 0.5M
Newfoundland 0.4M
Yukon 0.1M
Prince Edward I... 0.0M
Total Orders $ by Product Container
Small Box 5.1M
Jumbo Drum 3.3M
Jumbo Box 2.8M
Large Box 2.0M
Medium Box 0.7M
Small Pack 0.5M
Wrap Bag 0.3M
Total Orders $ by Customer Segment
1.9M (13.2%)
6.1M (41.1%)
2.8M (18.8%)
4.0M (26.9%)
Corporate
Consumer
Home Office
Small Business
Total Orders $ by Ship Mode
1.2M (8.0%)
6.1M (41.3%)
7.5M (50.7%)
Regular Air
Delivery Truck
Express Air
Order Amount by Product Category
6.1M
5.0M
3.6M
Technology
Furniture
Office Supplies
Product Category by Count
715
291
257
Office Supplies
Technology
Furniture
Total Orders $ by Month Name
1.4M
1.2M
1.3M
1.2M
1.2M
1.0M
1.1M
1.1M
1.3M
1.4M
1.2M
1.4M
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Order Amount by Region
West 6.4M
Prarie 2.7M
Ontario 2.3M
Atlantic 1.4M
Northwest Terr... 1.0M
Quebec 0.8M
Yukon 0.1M

# Repeat

- Color
- Visual Elements
- Fonts
- Ideas

# Alignment

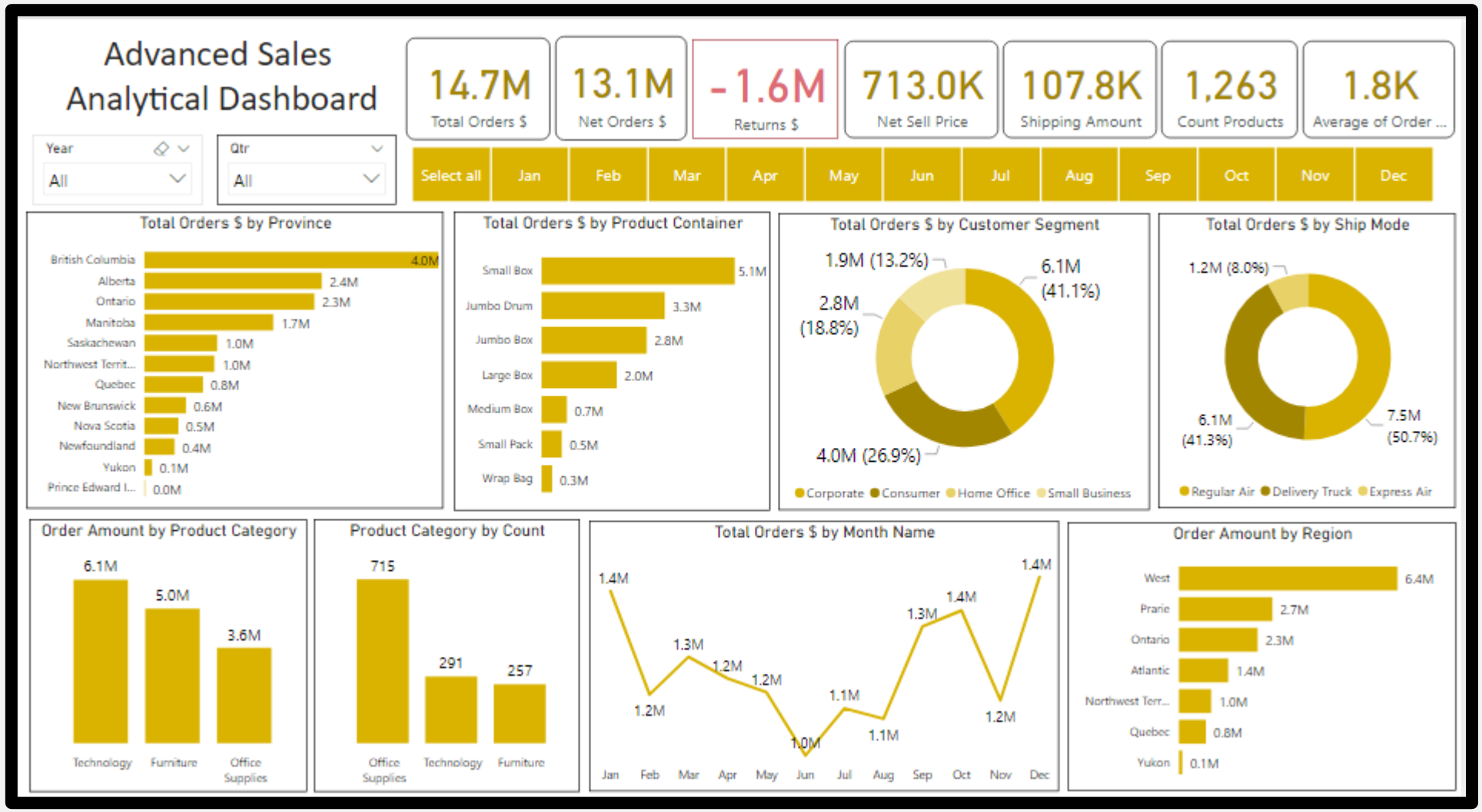


# Align

- Every edge of every shape with other shapes
- ALT Key
- Alignment tools

# Proximity

Proximity refers to visual search (i.e., your eyes scanning the page for information). In a dashboard, you're providing the tools and organizing them in a way that makes sense to users



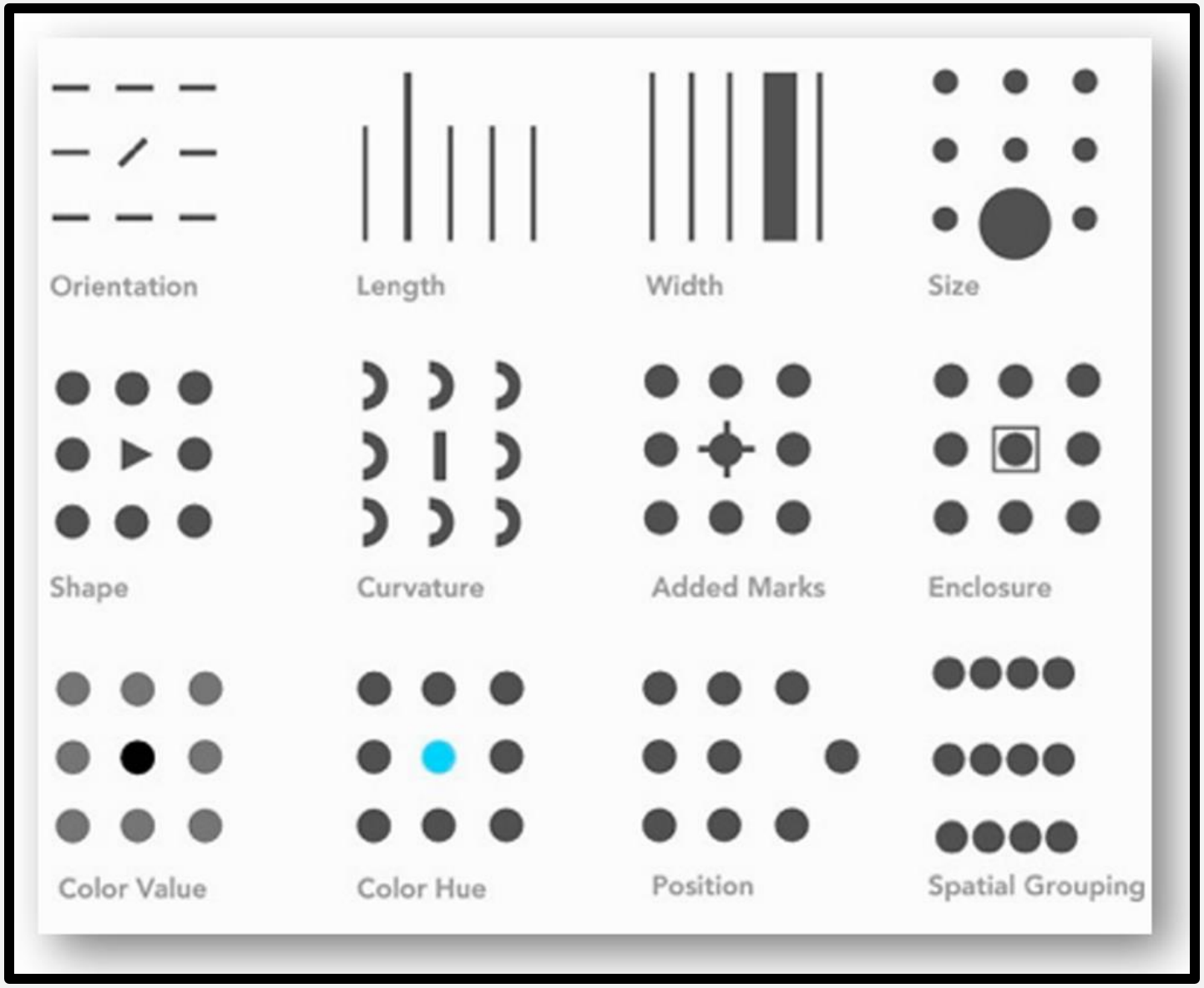
# Preattentive attributes:

They help us enable our audience to see what we want them to see before they even know they are seeing it.

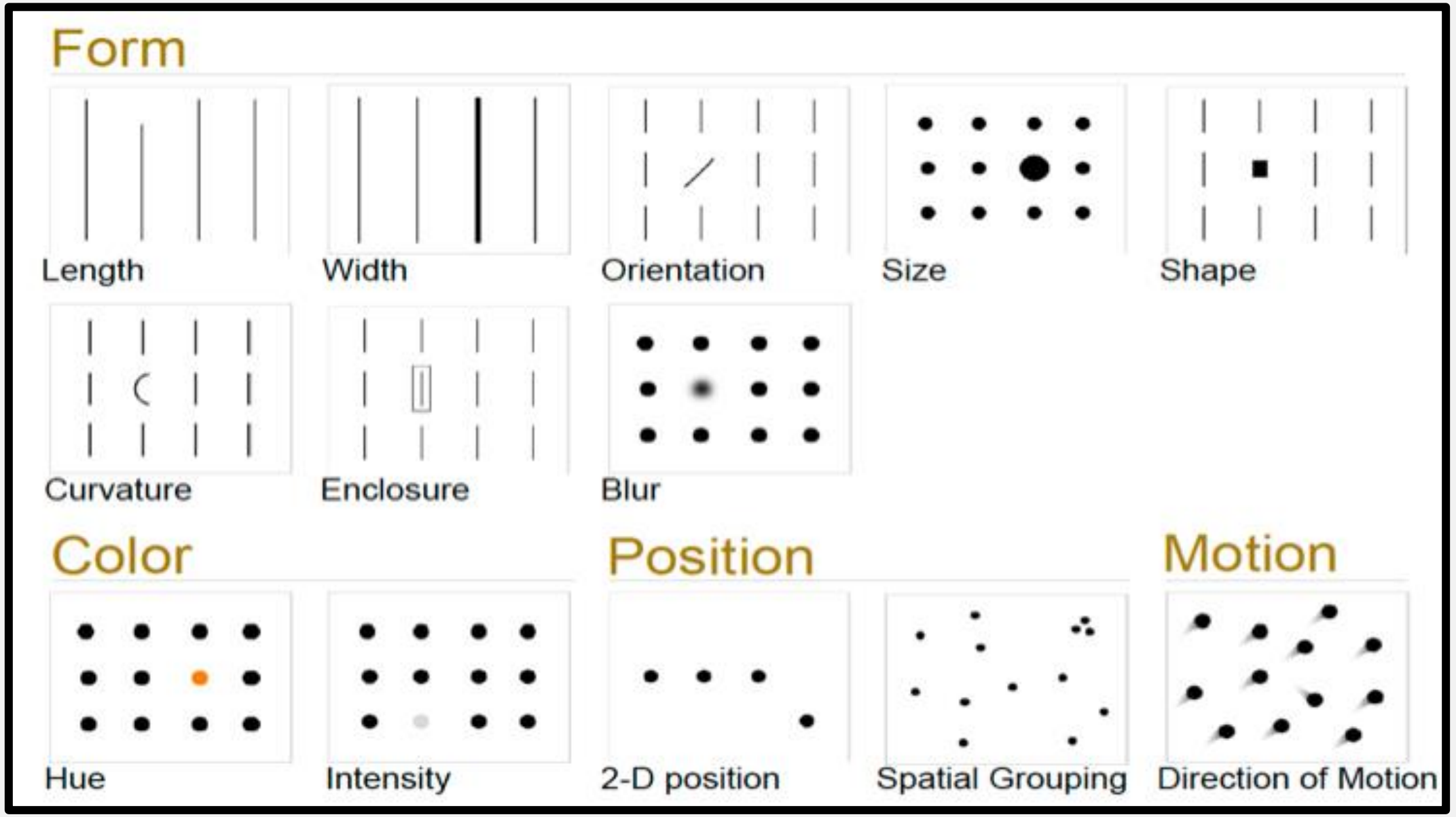
**Preattentive attributes** are hard-wired within our visual systems, and apply to just about everything that we can visually perceive.

This concept of preattentive visual properties can be applied to data visualizations in order to more effectively communicate what we want our data to show.

# Preattentive attributes



# Preattentive attributes



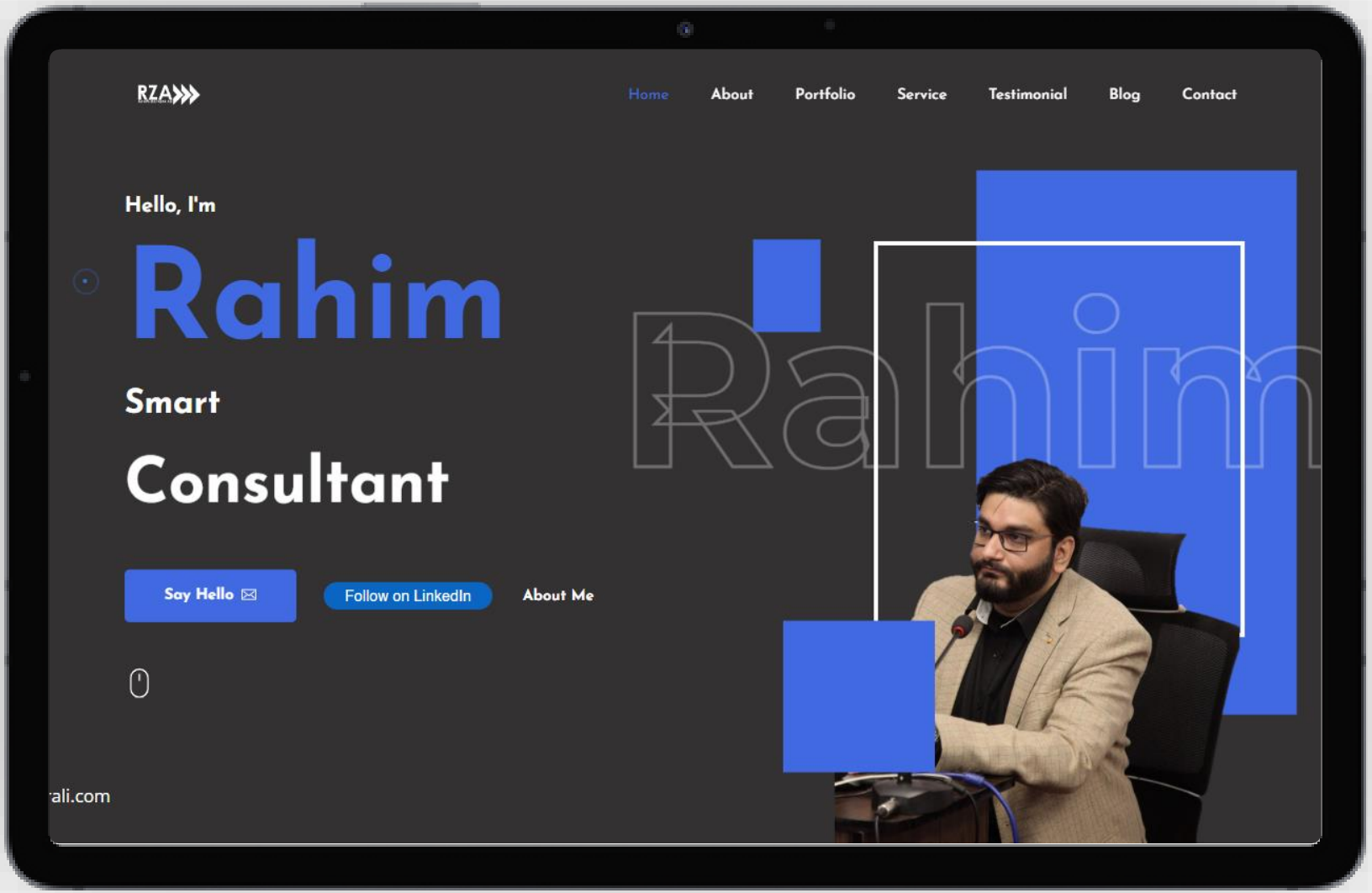
# Dashboard Design Rules



# 10 Design Rules / Suggestions!

1. Design with a specific goal in mind
2. Dashboard on a Single Page
3. Align the Elements (Slicers, Charts, Text Box)
4. Be Consistent with Design (Colors, Fonts, Charts)
5. Highlight the Most Relevant Information
6. Short the Numbers (Example 1.5M | 3.1B)
7. Choose the Right Colors (Theme)
8. Pick the Right Charts
9. Dashboard should present Data Storytelling for easy Decision Making
10. Show the Context (Actual vs Budget, Variances)

https://www.rahimzulfiqarali.com/



https://www.youtube.com/@ExcelBasement/

**Subscribe to my YouTube Channel “Excel Basement”**

